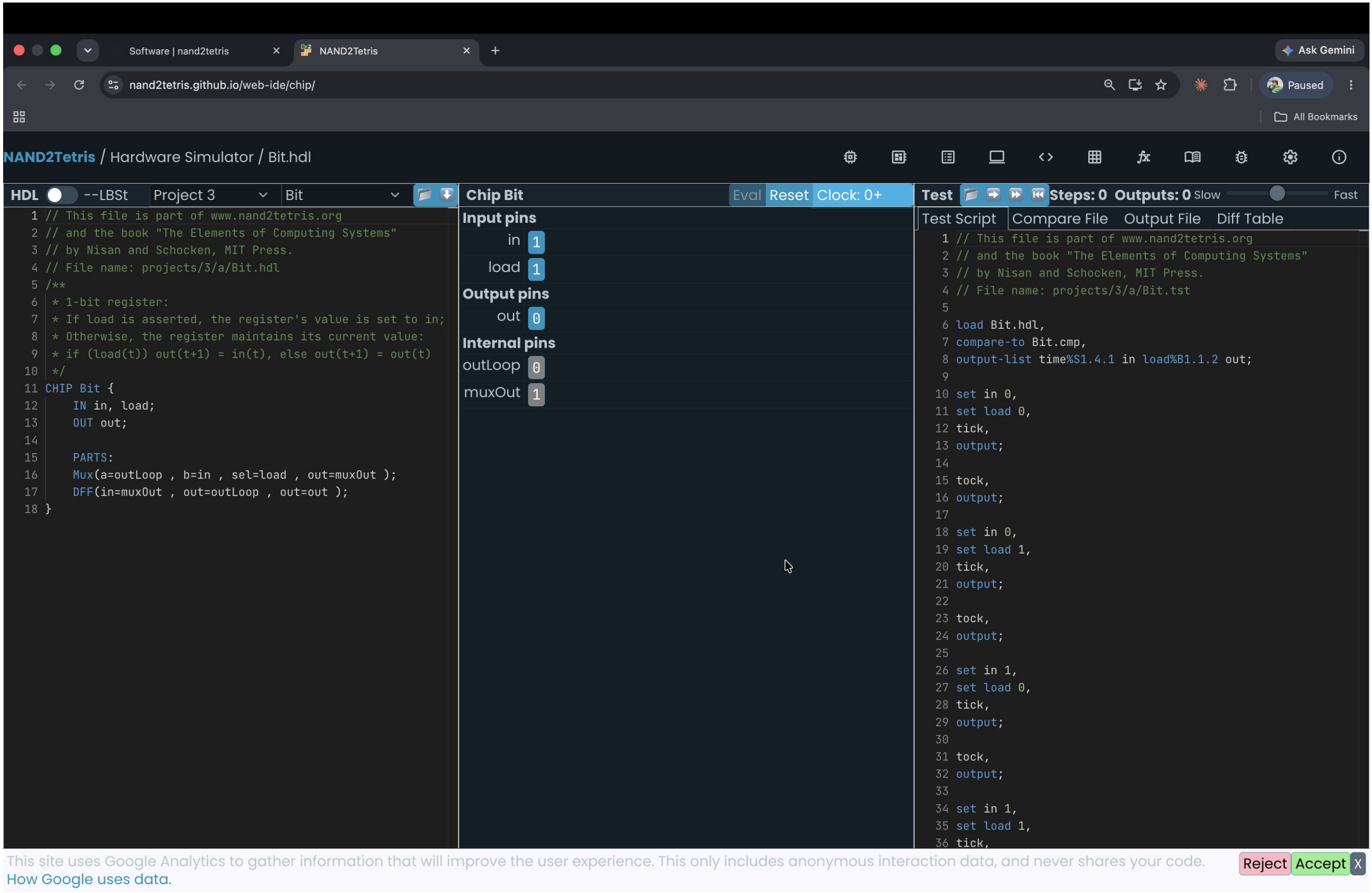
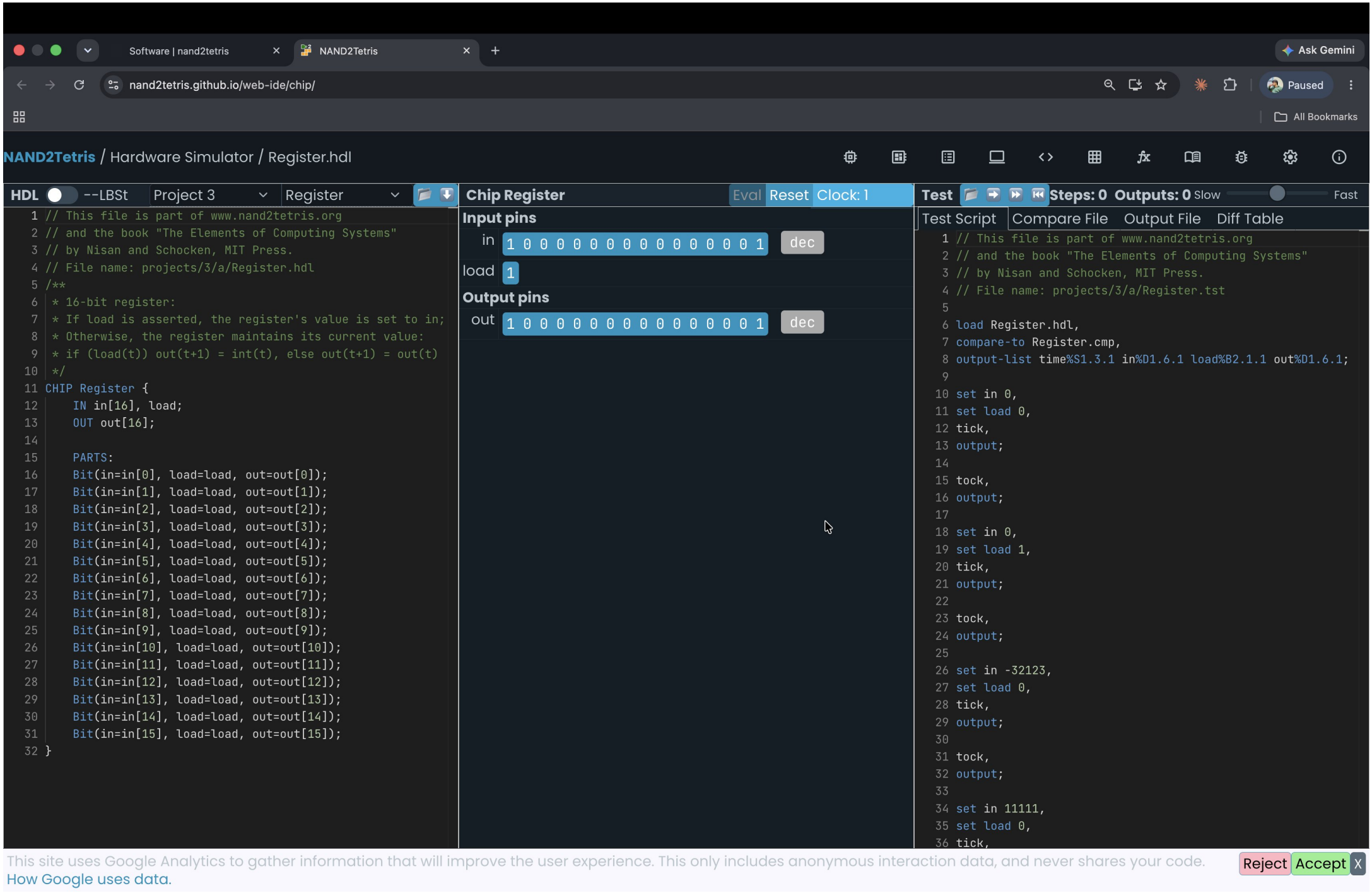
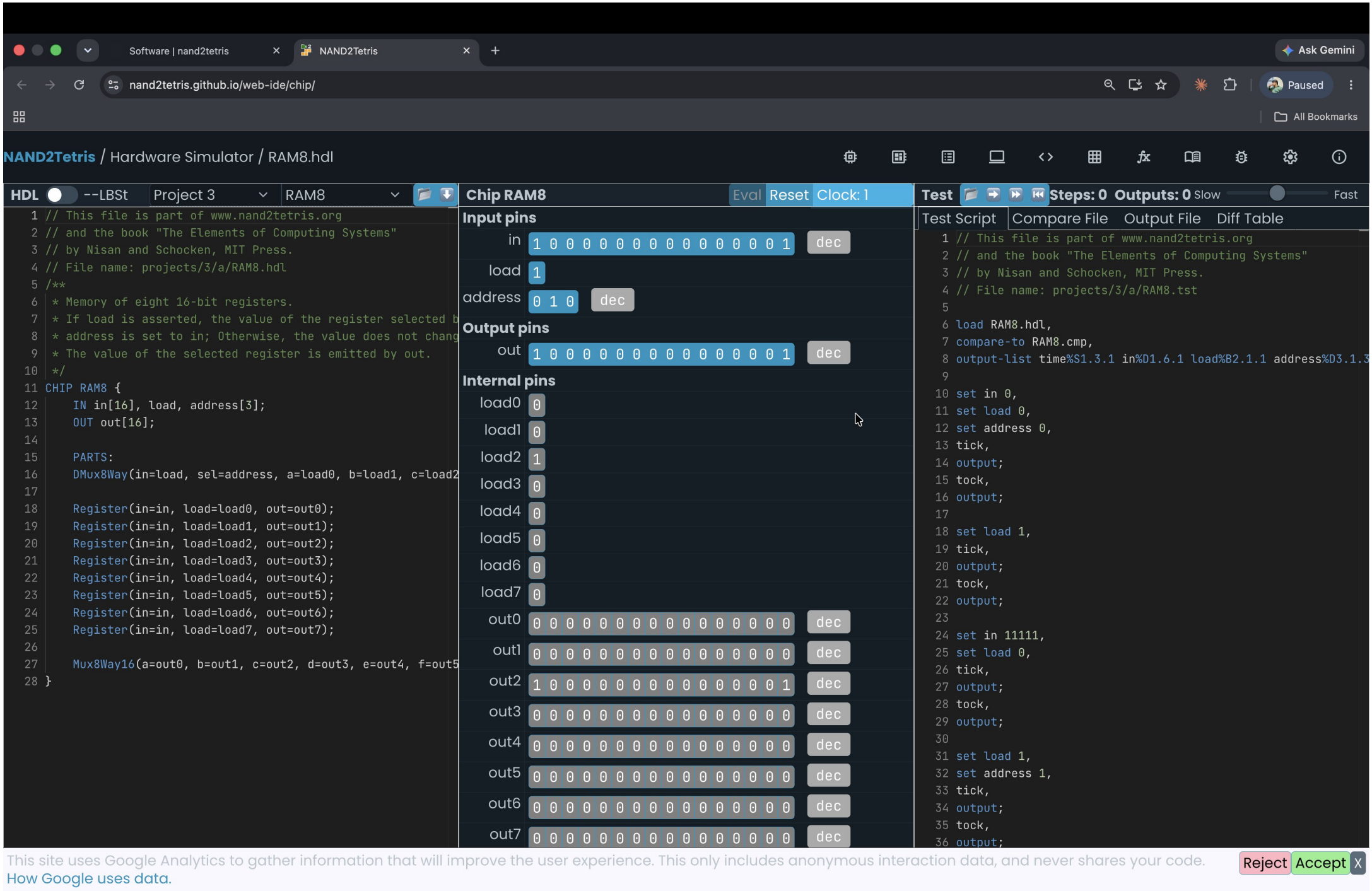
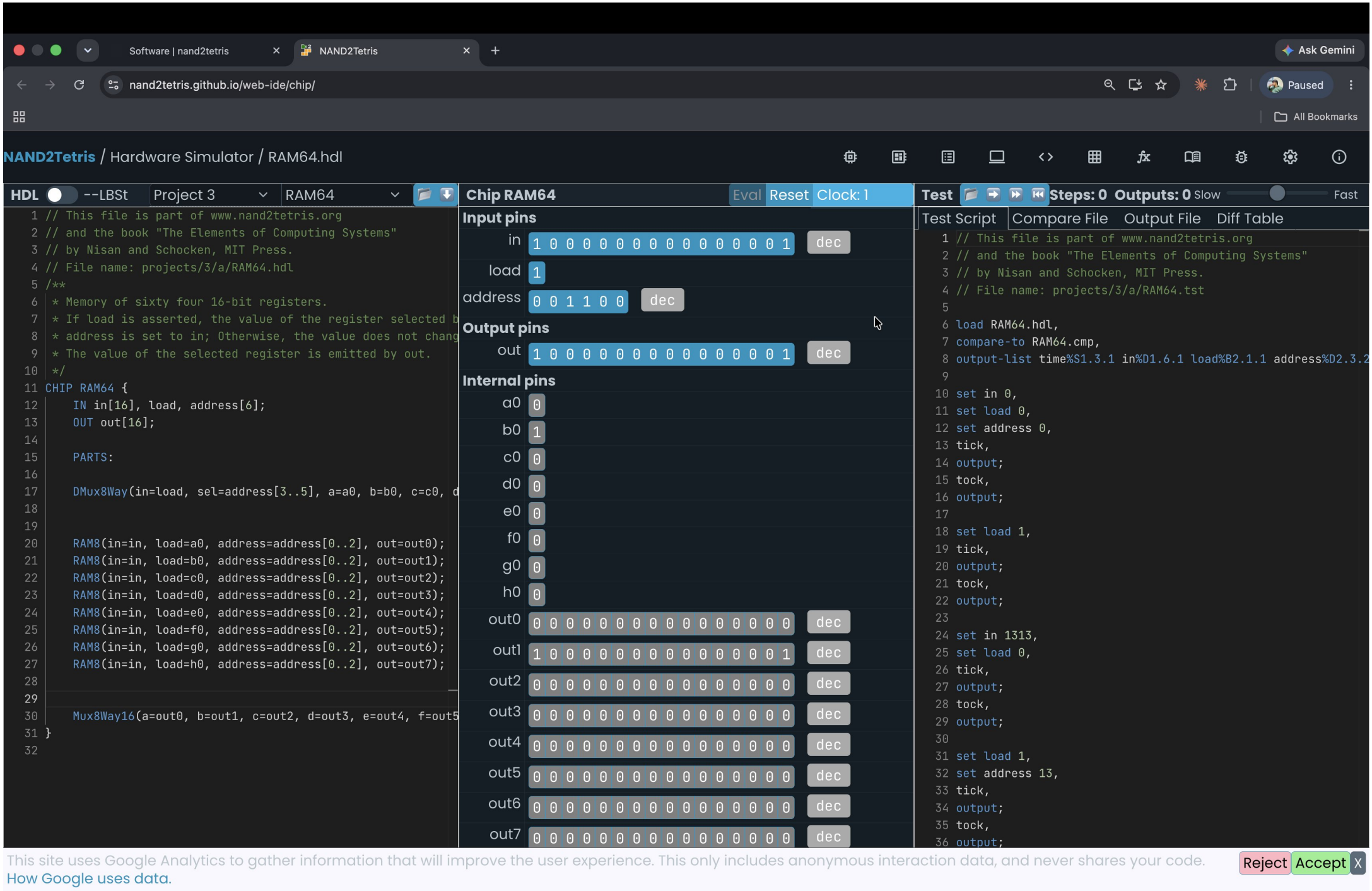
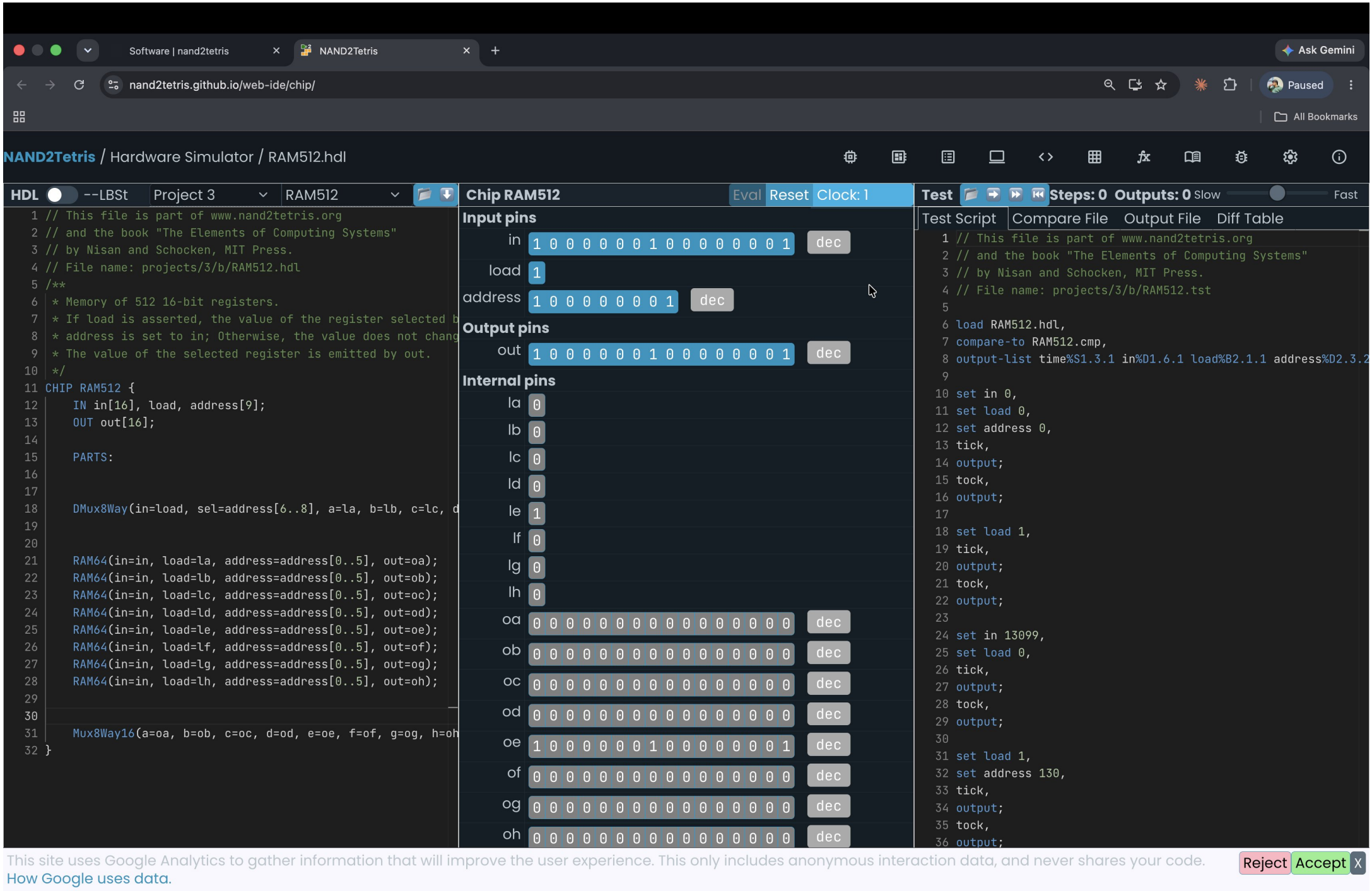
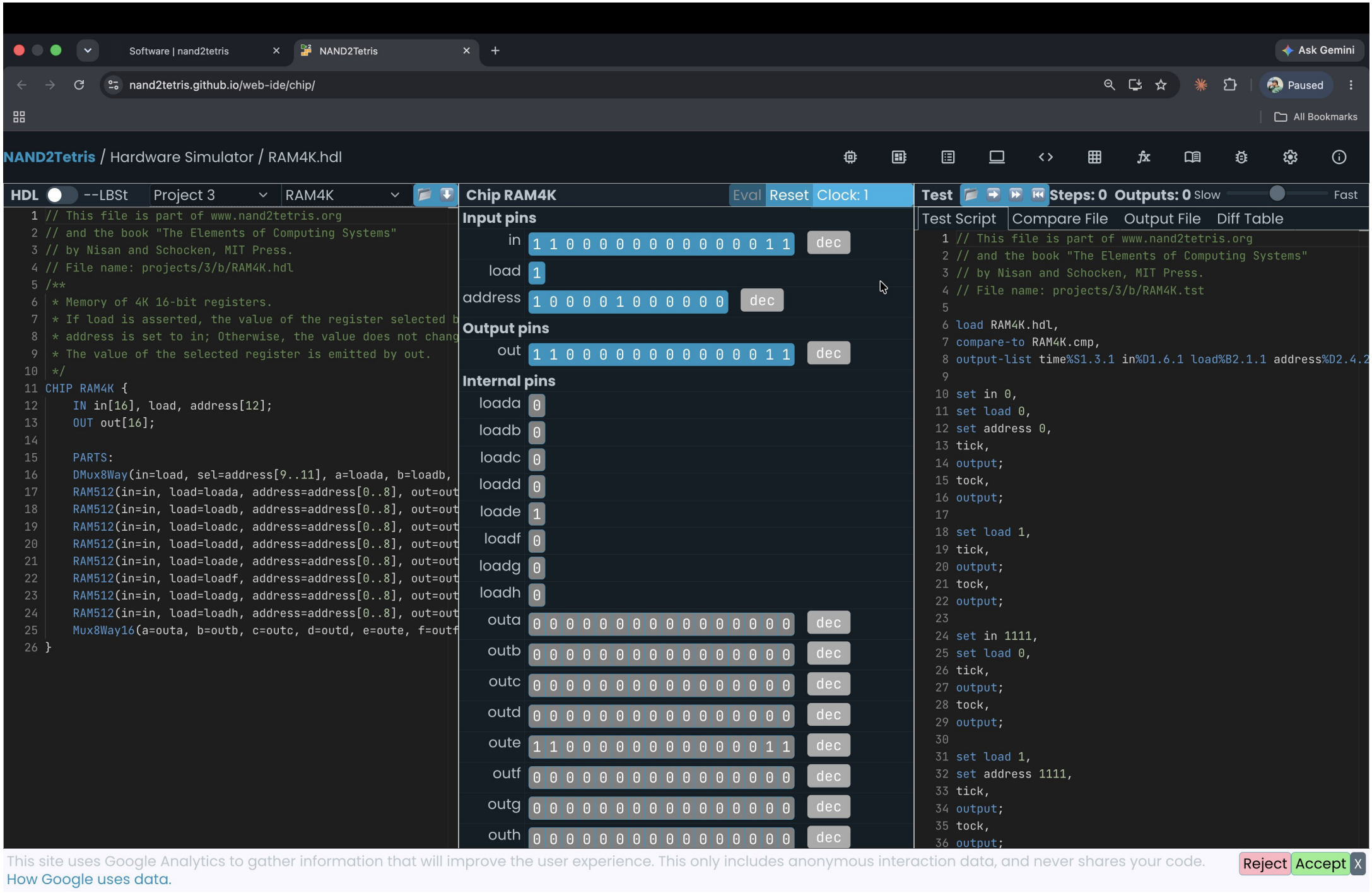

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NAND2Tetris / Hardware Simulator / Bit.hdl
HDL --LBSt Project 3 Bit
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/a/Bit.hdl
/**
* 1-bit register:
* If load is asserted, the register's value is set to in;
* Otherwise, the register maintains its current value:
* if (load(t)) out(t+1) = in(t), else out(t+1) = out(t)
*/
CHIP Bit {
IN in, load;
OUT out;
PARTS:
Mux(a=outLoop , b=in , sel=load , out=muxOut);
DFF(in=muxOut , out=outLoop , out=out);
}
Chip Bit Eval Reset Clock: 0+
Input pins
in 1
load 1
Output pins
out 0
Internal pins
outLoop 0
muxOut 1
Test Steps: 0 Outputs: 0 Slow Fast
Test Script Compare File Output File Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/a/Bit.tst
load Bit.hdl,
compare-to Bit.cmp,
output-list time%S1.4.1 in load%B1.1.2 out;
set in 0,
set load 0,
tick,
output;
tock,
output;
set in 0,
set load 1,
tick,
output;
tock,
output;
set in 1,
set load 0,
tick,
output;
tock,
output;
set in 1,
set load 1,
tick,
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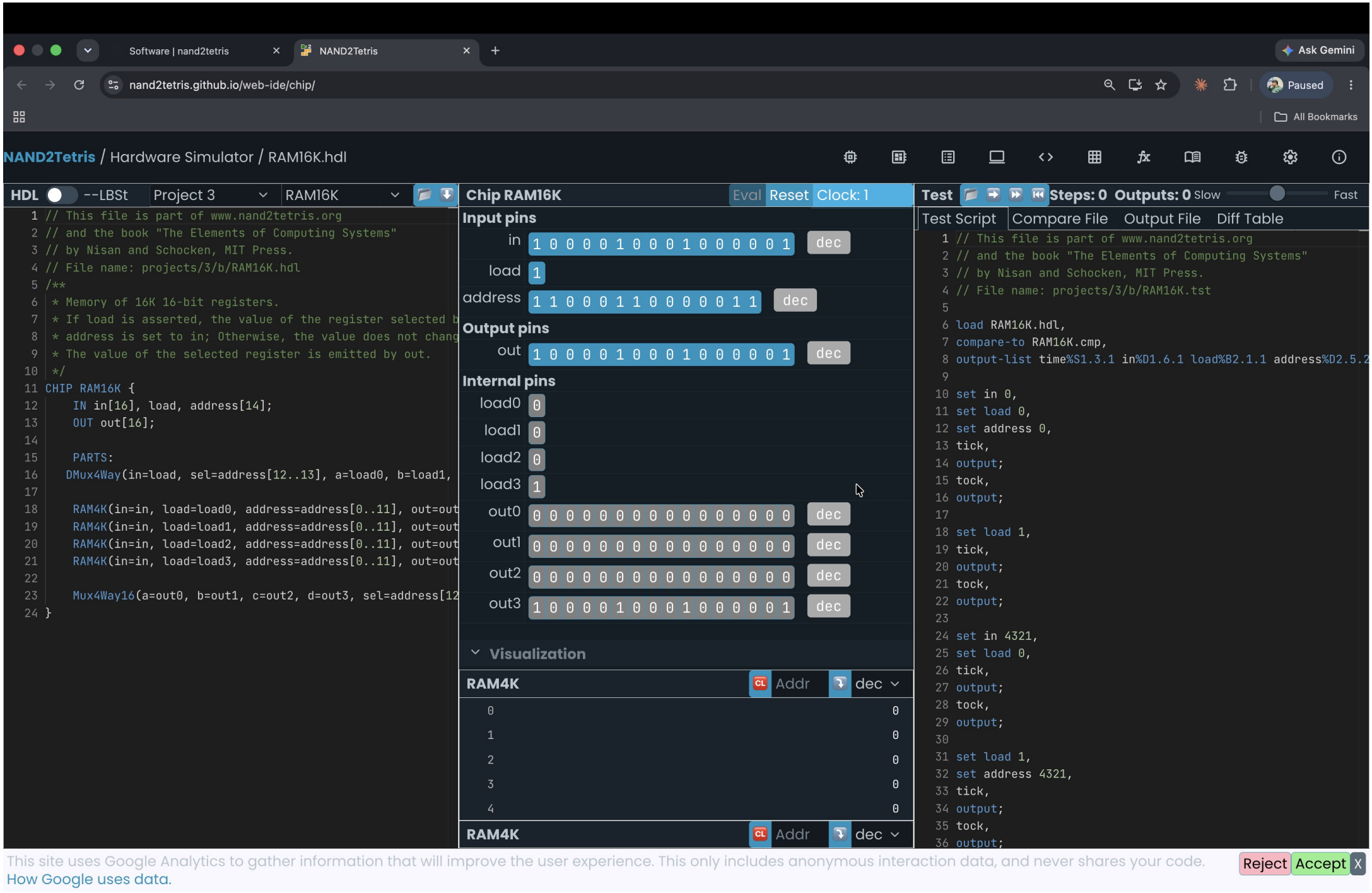

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NAND2Tetris / Hardware Simulator / Register.hdl
HDL --LBSt Project 3 Register
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/a/Register.hdl
/**
* 16-bit register:
* If load is asserted, the register's value is set to in;
* Otherwise, the register maintains its current value:
* if (load(t)) out(t+1) = int(t), else out(t+1) = out(t)
*/
CHIP Register {
IN in[16], load;
OUT out[16];
PARTS:
Bit(in=in[0], load=load, out=out[0]);
Bit(in=in[1], load=load, out=out[1]);
Bit(in=in[2], load=load, out=out[2]);
Bit(in=in[3], load=load, out=out[3]);
Bit(in=in[4], load=load, out=out[4]);
Bit(in=in[5], load=load, out=out[5]);
Bit(in=in[6], load=load, out=out[6]);
Bit(in=in[7], load=load, out=out[7]);
Bit(in=in[8], load=load, out=out[8]);
Bit(in=in[9], load=load, out=out[9]);
Bit(in=in[10], load=load, out=out[10]);
Bit(in=in[11], load=load, out=out[11]);
Bit(in=in[12], load=load, out=out[12]);
Bit(in=in[13], load=load, out=out[13]);
Bit(in=in[14], load=load, out=out[14]);
Bit(in=in[15], load=load, out=out[15]);
}
Chip Register
Eval
Reset
Clock: 1
Input pins
in 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1
dec
load 1
Output pins
out 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1
dec
Test
Steps: 0 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/a/Register.tst
load Register.hdl,
compare-to Register.cmp,
output-list time%S1.3.1 in%D1.6.1 load%B2.1.1 out%D1.6.1;
set in 0,
set load 0,
tick,
output;
tock,
output;
set in 0,
set load 1,
tick,
output;
tock,
output;
set in -32123,
set load 0,
tick,
output;
tock,
output;
set in 11111,
set load 0,
tick,
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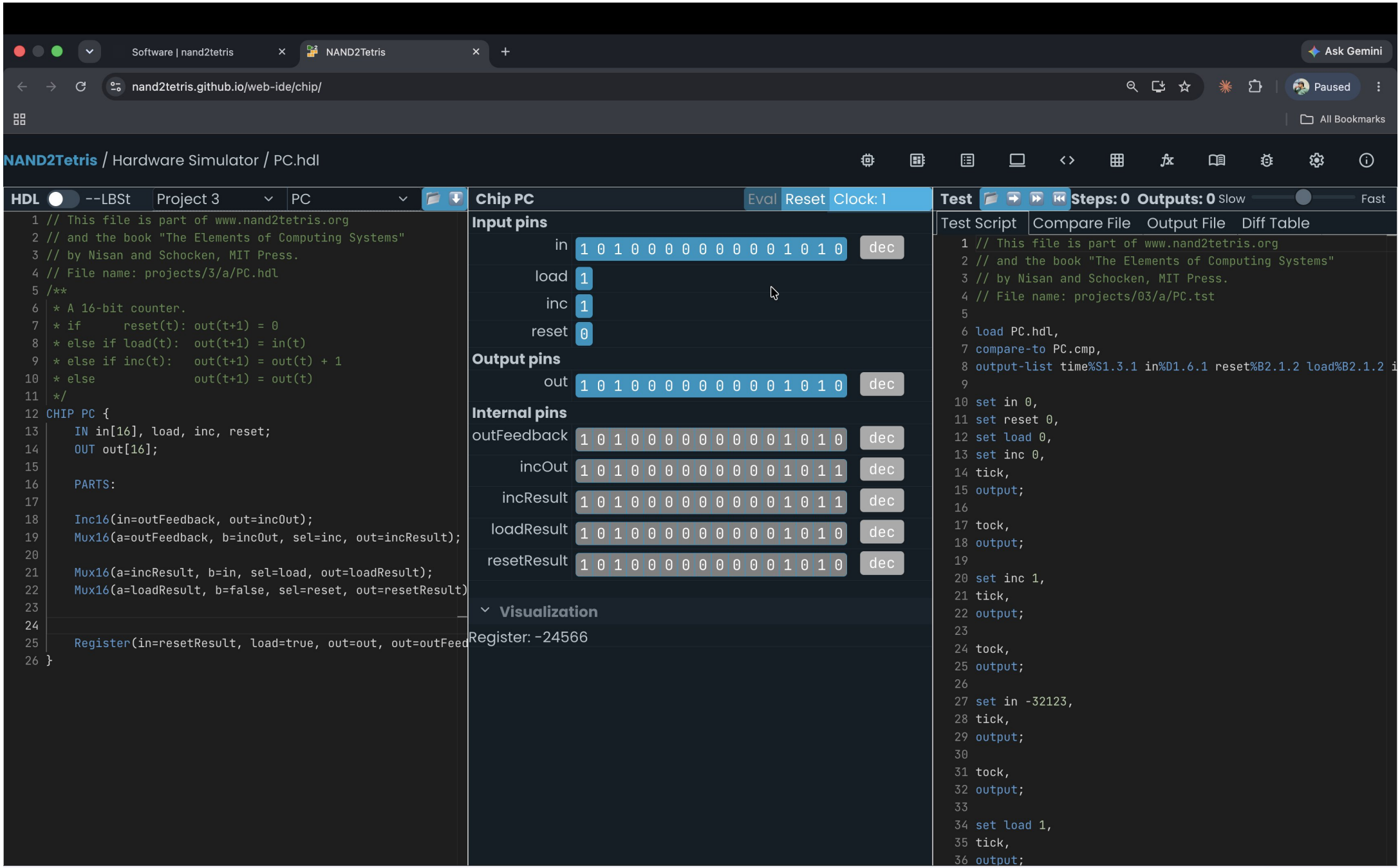

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NAND2Tetris / Hardware Simulator / RAM8.hdl
HDL --LBSt Project 3 RAM8
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/a/RAM8.hdl
/**
* Memory of eight 16-bit registers.
* If load is asserted, the value of the register selected b
* address is set to in; Otherwise, the value does not chang
* The value of the selected register is emitted by out.
*/
CHIP RAM8 {
IN in[16], load, address[3];
OUT out[16];
PARTS:
DMux8Way(in=load, sel=address, a=load0, b=load1, c=load2
Register(in=in, load=load0, out=out0);
Register(in=in, load=load1, out=out1);
Register(in=in, load=load2, out=out2);
Register(in=in, load=load3, out=out3);
Register(in=in, load=load4, out=out4);
Register(in=in, load=load5, out=out5);
Register(in=in, load=load6, out=out6);
Register(in=in, load=load7, out=out7);
Mux8Way16(a=out0, b=out1, c=out2, d=out3, e=out4, f=out5
}
Chip RAM8 Eval Reset Clock: 1
Input pins
in 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 dec
load 1
address 0 1 0 dec
Output pins
out 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 dec
Internal pins
load0 0
load1 0
load2 1
load3 0
load4 0
load5 0
load6 0
load7 0
out0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
out1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
out2 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 dec
out3 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
out4 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
out5 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
out6 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
out7 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
Test Steps: 0 Outputs: 0 Slow Fast
Test Script Compare File Output File Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/a/RAM8.tst
load RAM8.hdl,
compare-to RAM8.cmp,
output-list time%S1.3.1 in%D1.6.1 load%B2.1.1 address%D3.1.3
set in 0,
set load 0,
set address 0,
tick,
output;
tock,
output;
set load 1,
tick,
output;
tock,
output;
set in 11111,
set load 0,
tick,
output;
tock,
output;
set load 1,
set address 1,
tick,
output;
tock,
output;
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NAND2Tetris / Hardware Simulator / RAM64.hdl
HDL --LBSt Project 3 RAM64
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/a/RAM64.hdl
/**
* Memory of sixty four 16-bit registers.
* If load is asserted, the value of the register selected b
* address is set to in; Otherwise, the value does not chang
* The value of the selected register is emitted by out.
*/
CHIP RAM64 {
IN in[16], load, address[6];
OUT out[16];
PARTS:
DMux8Way(in=load, sel=address[3..5], a=a0, b=b0, c=c0, d
RAM8(in=in, load=a0, address=address[0..2], out=out0);
RAM8(in=in, load=b0, address=address[0..2], out=out1);
RAM8(in=in, load=c0, address=address[0..2], out=out2);
RAM8(in=in, load=d0, address=address[0..2], out=out3);
RAM8(in=in, load=e0, address=address[0..2], out=out4);
RAM8(in=in, load=f0, address=address[0..2], out=out5);
RAM8(in=in, load=g0, address=address[0..2], out=out6);
RAM8(in=in, load=h0, address=address[0..2], out=out7);
Mux8Way16(a=out0, b=out1, c=out2, d=out3, e=out4, f=out5
}
Chip RAM64 Eval Reset Clock: 1
Input pins
in 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 dec
load 1
address 0 0 1 1 0 0 dec
Output pins
out 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 dec
Internal pins
a0 0
b0 1
c0 0
d0 0
e0 0
f0 0
g0 0
h0 0
out0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
out1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 dec
out2 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
out3 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
out4 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
out5 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
out6 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
out7 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
Test Steps: 0 Outputs: 0 Slow Fast
Test Script Compare File Output File Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/a/RAM64.tst
load RAM64.hdl,
compare-to RAM64.cmp,
output-list time%S1.3.1 in%D1.6.1 load%B2.1.1 address%D2.3.2
set in 0,
set load 0,
set address 0,
tick,
output;
tock,
output;
set load 1,
tick,
output;
tock,
output;
set in 1313,
set load 0,
tick,
output;
tock,
output;
set load 1,
set address 13,
tick,
output;
tock,
output;
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NAND2Tetris / Hardware Simulator / RAM512.hdl
HDL --LBSt Project 3 RAM512
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/b/RAM512.hdl
/**
* Memory of 512 16-bit registers.
* If load is asserted, the value of the register selected b
* address is set to in; Otherwise, the value does not chang
* The value of the selected register is emitted by out.
*/
CHIP RAM512 {
IN in[16], load, address[9];
OUT out[16];
PARTS:
DMux8Way(in=load, sel=address[6..8], a=la, b=lb, c=lc, d
RAM64(in=in, load=la, address=address[0..5], out=oa);
RAM64(in=in, load=lb, address=address[0..5], out=ob);
RAM64(in=in, load=lc, address=address[0..5], out=oc);
RAM64(in=in, load=ld, address=address[0..5], out=od);
RAM64(in=in, load=le, address=address[0..5], out=oe);
RAM64(in=in, load=lf, address=address[0..5], out=of);
RAM64(in=in, load=lg, address=address[0..5], out=og);
RAM64(in=in, load=lh, address=address[0..5], out=oh);
Mux8Way16(a=oa, b=ob, c=oc, d=od, e=oe, f=of, g=og, h=oh
}
Chip RAM512 Eval Reset Clock: 1
Input pins
in 1 0 0 0 0 0 0 1 0 0 0 0 0 0 0 1 dec
load 1
address 1 0 0 0 0 0 0 0 1 dec
Output pins
out 1 0 0 0 0 0 0 1 0 0 0 0 0 0 0 1 dec
Internal pins
la 0
lb 0
lc 0
ld 0
le 1
lf 0
lg 0
lh 0
oa 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
ob 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
oc 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
od 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
oe 1 0 0 0 0 0 0 1 0 0 0 0 0 0 0 1 dec
of 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
og 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
oh 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
Test Steps: 0 Outputs: 0 Slow Fast
Test Script Compare File Output File Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/b/RAM512.tst
load RAM512.hdl,
compare-to RAM512.cmp,
output-list time%S1.3.1 in%D1.6.1 load%B2.1.1 address%D2.3.2
set in 0,
set load 0,
set address 0,
tick,
output;
tock,
output;
set load 1,
tick,
output;
tock,
output;
set in 13099,
set load 0,
tick,
output;
tock,
output;
set load 1,
set address 130,
tick,
output;
tock,
output;
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NAND2Tetris / Hardware Simulator / RAM4K.hdl
HDL --LBSt Project 3 RAM4K
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/b/RAM4K.hdl
/**
* Memory of 4K 16-bit registers.
* If load is asserted, the value of the register selected b
* address is set to in; Otherwise, the value does not chang
* The value of the selected register is emitted by out.
*/
CHIP RAM4K {
IN in[16], load, address[12];
OUT out[16];
PARTS:
DMux8Way(in=load, sel=address[9..11], a=loada, b=loadb,
RAM512(in=in, load=loada, address=address[0..8], out=out
RAM512(in=in, load=loadb, address=address[0..8], out=out
RAM512(in=in, load=loadc, address=address[0..8], out=out
RAM512(in=in, load=loadd, address=address[0..8], out=out
RAM512(in=in, load=loade, address=address[0..8], out=out
RAM512(in=in, load=loadf, address=address[0..8], out=out
RAM512(in=in, load=loadg, address=address[0..8], out=out
RAM512(in=in, load=loadh, address=address[0..8], out=out
Mux8Way16(a=outa, b=outb, c=outc, d=outd, e=oute, f=outf
}
Chip RAM4K Eval Reset Clock: 1
Input pins
in 1 1 0 0 0 0 0 0 0 0 0 0 0 0 1 1 dec
load 1
address 1 0 0 0 0 1 0 0 0 0 0 0 dec
Output pins
out 1 1 0 0 0 0 0 0 0 0 0 0 0 0 1 1 dec
Internal pins
loada 0
loadb 0
loadc 0
loadd 0
loade 1
loadf 0
loadg 0
loadh 0
outa 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
outb 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
outc 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
outd 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
oute 1 1 0 0 0 0 0 0 0 0 0 0 0 0 1 1 dec
outf 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
outg 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
outh 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
Test Steps: 0 Outputs: 0 Slow Fast
Test Script Compare File Output File Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/b/RAM4K.tst
load RAM4K.hdl,
compare-to RAM4K.cmp,
output-list time%S1.3.1 in%D1.6.1 load%B2.1.1 address%D2.4.2
set in 0,
set load 0,
set address 0,
tick,
output;
tock,
output;
set load 1,
tick,
output;
tock,
output;
set in 1111,
set load 0,
tick,
output;
tock,
output;
set load 1,
set address 1111,
tick,
output;
tock,
output;
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NAND2Tetris / Hardware Simulator / RAM16K.hdl
HDL --LBSt Project 3 RAM16K
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/b/RAM16K.hdl
/**
* Memory of 16K 16-bit registers.
* If load is asserted, the value of the register selected b
* address is set to in; Otherwise, the value does not chang
* The value of the selected register is emitted by out.
*/
CHIP RAM16K {
IN in[16], load, address[14];
OUT out[16];
PARTS:
DMux4Way(in=load, sel=address[12..13], a=load0, b=load1,
RAM4K(in=in, load=load0, address=address[0..11], out=out
RAM4K(in=in, load=load1, address=address[0..11], out=out
RAM4K(in=in, load=load2, address=address[0..11], out=out
RAM4K(in=in, load=load3, address=address[0..11], out=out
Mux4Way16(a=out0, b=out1, c=out2, d=out3, sel=address[12
}
Chip RAM16K Eval Reset Clock: 1
Input pins
in 1 0 0 0 0 1 0 0 0 1 0 0 0 0 0 1 dec
load 1
address 1 1 0 0 0 1 1 0 0 0 0 0 1 1 dec
Output pins
out 1 0 0 0 0 1 0 0 0 1 0 0 0 0 0 1 dec
Internal pins
load0 0
load1 0
load2 0
load3 1
out0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
out1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
out2 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
out3 1 0 0 0 0 1 0 0 0 1 0 0 0 0 0 1 dec
Visualization
RAM4K CL Addr dec
0 0
1 0
2 0
3 0
4 0
RAM4K CL Addr dec
Test Steps: 0 Outputs: 0 Slow Fast
Test Script Compare File Output File Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/b/RAM16K.tst
load RAM16K.hdl,
compare-to RAM16K.cmp,
output-list time%S1.3.1 in%D1.6.1 load%B2.1.1 address%D2.5.2
set in 0,
set load 0,
set address 0,
tick,
output;
tock,
output;
set load 1,
tick,
output;
tock,
output;
set in 4321,
set load 0,
tick,
output;
tock,
output;
set load 1,
set address 4321,
tick,
output;
tock,
output;
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NAND2Tetris / Hardware Simulator / PC.hdl
HDL --LBSt Project 3 PC
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/a/PC.hdl
/**
* A 16-bit counter.
* if reset(t): out(t+1) = 0
* else if load(t): out(t+1) = in(t)
* else if inc(t): out(t+1) = out(t) + 1
* else out(t+1) = out(t)
*/
CHIP PC {
IN in[16], load, inc, reset;
OUT out[16];
PARTS:
Inc16(in=outFeedback, out=incOut);
Mux16(a=outFeedback, b=incOut, sel=inc, out=incResult);
Mux16(a=incResult, b=in, sel=load, out=loadResult);
Mux16(a=loadResult, b=false, sel=reset, out=resetResult)
Register(in=resetResult, load=true, out=out, out=outFeed
}
Chip PC
Eval
Reset
Clock: 1
Input pins
in 1 0 1 0 0 0 0 0 0 0 0 0 1 0 1 0 dec
load 1
inc 1
reset 0
Output pins
out 1 0 1 0 0 0 0 0 0 0 0 0 1 0 1 0 dec
Internal pins
outFeedback 1 0 1 0 0 0 0 0 0 0 0 0 1 0 1 0 dec
incOut 1 0 1 0 0 0 0 0 0 0 0 0 1 0 1 1 dec
incResult 1 0 1 0 0 0 0 0 0 0 0 0 1 0 1 1 dec
loadResult 1 0 1 0 0 0 0 0 0 0 0 0 1 0 1 0 dec
resetResult 1 0 1 0 0 0 0 0 0 0 0 0 1 0 1 0 dec
Visualization
Register: -24566
Test
Steps: 0 Outputs: 0 Slow Fast
Test Script
Compare File
Output File
Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/03/a/PC.tst
load PC.hdl,
compare-to PC.cmp,
output-list time%S1.3.1 in%D1.6.1 reset%B2.1.2 load%B2.1.2 i
set in 0,
set reset 0,
set load 0,
set inc 0,
tick,
output;
tock,
output;
set inc 1,
tick,
output;
tock,
output;
set in -32123,
tick,
output;
tock,
output;
set load 1,
tick,
output;
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